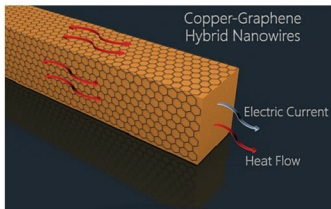


Energy storage. Being able to create supercapacitors out of graphene will possibly be the biggest step in electronics engineering. While development of electronic components has been progressing at a very high rate over the last 20 years, power storage solutions such as batteries and capacitors have been the primary limiting factor due to size, power capacity and efficiency (most types of batteries are very inefficient, and capacitors are even less so).

For example, with the development of currently-available lithium-ion batteries, it is difficult to create a balance between energy density and power density. In this situation, it is essentially about compromising one for the other.

Graphene is also being used to boost not only the capacity and charge rate of batteries but also longevity. Currently, while such materials as silicone are able to store large amounts of energy, that potential amount diminishes drastically on every charge or recharge. With graphene tin-oxide being used as an anode in lithium-ion batteries, for example, batteries can be made to last much longer between charges (potential capacity has increased by a factor of ten). And with almost no reduction in storage capacity between charges, it would effectively make electronically-powered vehicles a much more viable transport solution in the future.

This means that batteries (or capacitors) can be developed to last much longer and at higher capacities than previously realised. Also, it means that electronic devices may be able to be charged within seconds, rather than minutes or hours, and have hugely-improved longevity.



This illustration depicts a copper nanowire coated with graphene—an ultrathin layer of carbon—which lowers resistance and heating, suggesting potential applications in computer chips and flexible displays (Image courtesy: www.purdue.edu)

Textile electrodes. A team of researchers has used chemical vapour deposition to fabricate monolayer graphene. This research has developed a way to peel graphene off the copper sheet and transfer it to a yarn without compromising the electronic properties of graphene. The methodology that has been developed to prepare transparent and conductive textile fibres by coating these with graphene will now open the way to the integration of electronic devices on these textile fibres.

Researchers envision this process will enable a range of potential applications including textile GPS systems, biomedical monitoring, personal security or even communication tools for those who are sensory impaired.

Graphene-coated copper nanowires. In research that sort of bridges flexible displays with interconnects, researchers have been successful in developing copper nanowires coated with graphene to lower resistance and susceptibility to heating of copper wires. This research could allow copper wires to be used in a range of electronics including the flexible variety.

Highly-conductive copper nanowires are essential for efficient data transfer and heat conduc-

tion in many applications like high-performance semiconductor chips and transparent displays. This is compelling evidence for improved speed and thermal management by adapting the copper-graphene hybrid technology in future silicon chips and flexible electronics applications.

Graphene in thermal management. As circuit densities and clock speeds of chips are steadily rising, thermal-management

issues are becoming increasingly paramount. Because of graphene's high thermal conductivity, it has been investigated as a potential solution to some of these thermal-management issues.

Now, researchers have brought that promise of graphene one step closer with the discovery that, heat propagates in the form of a wave, just like sound in air. This is extremely-valuable information for engineers, who could adapt the design of future electronic components using some of these novel 2D materials' properties. The work should help engineers better understand the mechanisms of thermal conductivity in graphene and other 2D materials, and become a valuable tool for those who are looking to use graphene for thermal-management solutions.

Magnetised graphene. While the over-riding preoccupation with graphene in electronics may be to see it used in digital logic applications, it may provide enormous improvements to today's digital memory solutions. Graphene has been imbued with magnetic properties that could lead to a million-fold increase in capacity over today's hard drives. While some research teams have managed to magnetise graphene, others were able to achieve it with a relatively-simple